Single-Index Bandits

Bandits whose reward is an unknown function of an unknown projection

CS6007 project notes

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What this folder is about. All five papers study rewards of the form $y = f(\langle x, \theta^* \rangle) + \eta$ (or an extension) where *both* the direction θ^* and the link f are unknown, so a linear or GLM bandit would be misspecified. They are all **single-agent** (one is batched). Our whole project extends this line to many agents, so this folder is our base: ZoomSIB-UCB is the algorithm we federate and decentralise.

Common machinery. Every paper estimates θ^* with a *Stein-type* estimator: for contexts with density p and score $S(x) = -\nabla \log p(x)$, Stein's identity gives $\mathbb{E}[y\, S(x)] = \mu^* \theta^*$ with $\mu^* = \mathbb{E}[f'(\langle x, \theta^* \rangle)]$. This needs $\mu^* \neq 0$, an assumption that recurs below as an open problem.

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1 Dey, Bhore & Ghosh (2026): ZoomSIB-UCB, optimal regret

D. Dey, S. Bhore, A. Ghosh. Optimal Regret for Single Index Bandits. arXiv:2605.09454, 2026.
File: `2605.09454-Dey-Bhore-Ghosh-Optimal-Regret-Single-Index-Bandits.pdf`

Setting.

- One agent; K i.i.d. contexts per round; f bounded and Lipschitz, possibly *non-monotone*; $\|\theta^*\|_1 = 1$.
- Assumes $\mu^* \geq c_0/\text{polylog}$ (bounded away from 0).

What they did.

- **Phase 1:** explore uniformly; estimate θ^* by a normalised, truncated Stein estimator.
- **Phase 2:** project arms onto $\hat{\theta}$, cut the index line into $\sim T^{1/3}$ bins, and run *sleeping* UCB over the bins available each round.
- Experiments use an adaptive stopping rule for Phase 1; real-data tests on KDD Cup 99 and Forest Cover.

Main results.

- Regret $\tilde{O}(T^{2/3})$ ($\tilde{O}(d^2T^{2/3})$ with dimension), improving GSTOR's $\tilde{O}(T^{3/4})$.
- A matching minimax lower bound $\tilde{\Omega}(T^{2/3})$ for non-monotone links (built in $d = 1$).
- Sample splitting: $\hat{\theta}_0$ uses only Phase-1 data, so Phase 2's martingale argument holds.

Why it matters for our project.

- Our base algorithm. Fed-ZoomSIB and Dec-ZoomSIB-Relay federate its two phases; our server proofs reuse its Stein lemma and lower bound.
- E29 tests exactly its $\mu^* \neq 0$ assumption across heterogeneous agents.

Open problems.

- **Stated by the authors:** Remove the requirement $\mu^* \neq 0$ ("can an alternative technique be developed?").
- **Stated by the authors:** Go beyond bounded Lipschitz f.
- **Stated by the authors:** The multi-index setting $y = f(B^\top x)$.
- **Our reading:** The grid is fixed (one-shot binning), so the rate is $T^{2/3}$ on every instance; truly adaptive (zooming) bins could give instance-dependent rates. Our E3 measured exponents 0.29–0.60.

2 Kang et al. (2025/ICLR 2026): introducing single-index bandits

Y. Kang, M. Liu, B. Yi, J. Lyu, Z. Zhang, D. Zhou, Y. Li. Single Index Bandits: Generalized Linear Contextual Bandits with Unknown Reward Functions. ICLR 2026 (arXiv:2506.12751).
File: `arxiv-2506.12751.pdf`

Setting.

- GLM bandits where the link is unknown ("single index bandits"). Motivation: a misspecified link can give linear regret for every existing GLM method.

What they did.

- **STOR** and **ESTOR** for *monotone* f: Stein-based estimation of θ^*, then explore-then-commit.

- A sparse high-dimensional extension.
- **GSTOR** for general f under Gaussian design: double exploration, kernel regression for f.
- Baselines: LinUCB, LinTS, UCB-GLM, GLM-TSL, including deliberately misspecified versions.

Main results.

- ESTOR: nearly optimal $\tilde{O}(\sqrt{T})$ for monotone f; the same rate with sparsity in high dimension.
- GSTOR: sublinear regret for general f ($\tilde{O}(T^{3/4})$, later improved to $T^{2/3}$ by Dey et al.).

Why it matters for our project.

- First paper of the area; source of the truncated Stein estimator and of ESTOR/GSTOR, which are our single-agent baselines.

Open problems.

- **Our reading:** The $T^{3/4}$ rate for general f was not optimal (resolved by Dey et al.).
- **Our reading:** GSTOR's guarantees use a Gaussian design; its double exploration is costly in practice (Dey et al. report 79–90% exploration on real data).

3 Arya, Bhattacharjee & Sriperumbudur (2026): kernel single-index bandits with inference

S. Arya, S. Bhattacharjee, B. K. Sriperumbudur. Kernel Single-Index Bandits: Estimation, Inference, and Learning. arXiv:2603.18938, 2026.
File: `arxiv-2603.18938.pdf`

Setting.

- Finitely many arms, each with its *own* index parameter, and an unknown link; covariates evolve under the bandit policy (adaptively collected data).

What they did.

- A kernelised ε-greedy policy: Stein-based index estimation plus inverse-propensity-weighted kernel ridge regression for the reward functions.

Main results.

- Asymptotic normality of the index estimator under adaptive sampling (valid confidence regions).
- A directional functional CLT in an RKHS for the link estimate (pointwise confidence intervals).
- Finite-time regret, including $\tilde{O}(\sqrt{T})$ under common-link Lipschitz conditions.

Why it matters for our project.

- Different goal (statistical inference, not optimal regret) and a different model (arm-specific indices). Related work, not a baseline.

Open problems.

- **Stated by the authors:** Multi-index or partially linear reward structures.
- **Stated by the authors:** Uniform or anytime-valid confidence bands for sequential monitoring.
- **Stated by the authors:** Robustness to model misspecification.
- **Stated by the authors:** Exploration strategies that explicitly preserve inferential guarantees.

4 Liu, Kang & Lee (Sep 2026): low-rank single-index bandits

Z. Liu, Y. Kang, T. C. M. Lee. Low-Rank Single-Index Bandits with Unknown Links: From Matrices to Tensors. arXiv:2609.33025, 2026.
File: 2609.33025v1.pdf

Setting.

- Matrix and tensor bandits with a *low-rank* index parameter and an unknown shared Lipschitz link; known regular candidate distributions; finite-variance noise.

What they did.

- **T-ESTOR** for monotone links: robust, rank-adaptive Stein estimation plus epoch-based greedy selection.
- **T-BSTOR** for non-monotone links: structured estimation plus robust bin-based learning (ZoomSIB-like).

Main results.

- Monotone: square-root regret with dimension dependence set by the rank, and a matching lower bound (rank, dimension, horizon) at large horizons.
- Non-monotone: the optimal $\tilde{O}(T^{2/3})$ horizon rate for fixed dimensions.

Why it matters for our project.

- Shows the Kang group is actively extending single-index bandits; a scooping-risk signal for multi-agent work.
- Independent confirmation that "zero first-order Stein signal" ($\mu^* = 0$) is open, the case E29 studies.

Open problems.

- **Stated by the authors:** Certify learned (not exactly known) selected scores.
- **Stated by the authors:** **Handle zero first-order signals** ($\mu^* = 0$).
- **Stated by the authors:** Guarantees for the adaptive exploration used in their experiments.

5 Arya & Song (2025): batched single-index bandits

S. Arya, H. Song. Batched Single-Index Global Multi-Armed Bandits with Covariates. arXiv:2503.00565, 2025.

File: 2503.00565v3.pdf

Setting.

- K arms with rewards related through a shared single-index direction; feedback arrives in M batches (policy updated only between batches).

What they did.

- **BIDS**: batched successive arm elimination with *dynamic binning* along the index direction.
- Two regimes: a pilot direction is given, or the direction is estimated from data.

Main results.

- A lower bound for the K-armed problem, and an upper bound matching it (fixed K) given an accurate pilot direction.
- With $M \gtrsim \log\log T$ batches the bound simplifies to $(K \log T)^{(1+\alpha)/3} T^{1-(1+\alpha)/3}$, avoiding the curse of dimensionality (a $d = 1$ rate).

Why it matters for our project.

- The closest single-index work to *limited communication*: a batch is like a sync round. Relevant to our E27, where fixed batch schedules did *not* match per-round sync for our UCB agents.

Open problems.

- **Stated by the authors:** The upper bound depends polynomially on K while the lower bound is only logarithmic in K (Remark 4.3).
- **Stated by the authors:** Without a pilot, matching rates hold only when the margin parameter and batch size lie in restricted ranges.

6 Open problems across this folder

- $\mu^* = 0$ **is open in two papers** (Dey et al.; Liu–Kang–Lee). Second-order Stein (our E25) and multi-agent pooling (our E29) are two routes.
- **Multi-index** rewards are open (Dey et al.; Arya et al.).
- **No multi-agent single-index work exists** in this folder; batching (Arya & Song) is the closest communication-limited setting.
- **Adaptivity**: a fixed grid gives worst-case $T^{2/3}$; instance-dependent (zooming) rates are not studied.